David G Courtney BSc PhD

Senior Lecturer in Molecular Virology, Queen's University Belfast
UK Ambassador for the European Research Council

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Molecular virologist and RNA biologist. David leads The Courtney Group at Queen's University Belfast, which studies how influenza A virus RNA recruits host machinery to be modified, exported, translated and trafficked, and how those dependencies can be turned into RNA-based therapeutics. He has won over €1.8M in competitive funding as sole PI, sits on BBSRC and European funding panels, and represents the United Kingdom as an Ambassador for the European Research Council.

1,428 citations · h-index 19 · 47 indexed works (OpenAlex, 2026-10-12)

APPOINTMENTS

2025 – **Senior Lecturer in Molecular Virology**, Queen's University Belfast, UK

2021 – 2025 **Lecturer and ERC Research Fellow**, Queen's University Belfast, UK
Founded The Courtney Group with an ERC Starting Grant.

2020 – 2021 **Postdoctoral Research Fellow in Molecular Virology**, Queen's University Belfast, UK
High-throughput drug repurposing screens against SARS-CoV-2 in airway epithelial cells.

2019 – 2020 **Marie Skłodowska-Curie Postdoctoral Fellow**, Institut Pasteur, Paris, France
Developed methods to identify the host RNA-binding proteins of influenza A virus RNAs.

2016 – 2019 **Marie Skłodowska-Curie Postdoctoral Fellow**, Duke University, North Carolina, USA
Showed how m6A and m5C RNA modifications regulate influenza A virus, HIV-1 and other viruses.

2010 – 2011 **Industrial placement, BioPharm R&D**, GlaxoSmithKline, UK
Twelve months in molecular expression and cell line development.

EDUCATION

2012 – 2015 **PhD, Molecular Biology**, Ulster University, UK
The development of personalised medicines for corneal dystrophies (siRNA and CRISPR therapeutics).

2008 – 2012 **BSc (Hons) Biomedical Sciences with industrial studies, First Class**, Ulster University, UK

LEADERSHIP AND SERVICE

2026 – **Entrepreneurial Lead, Innovate UK ICURe Discover programme**, Innovate UK, UK Research and Innovation
Competitively selected for the UKRI-funded programme that helps researchers test the commercial potential of their research through market discovery.

2025 – **UK Ambassador for the European Research Council**, Ambassadors for the ERC network
One of 32 ERC grantees chosen to make the case for frontier research to policymakers, the media and the research community.

Current **Vice-Chair, Marie Skłodowska-Curie Actions Postdoctoral Fellowships panel**, European Commission

2024 – **Member, BBSRC Panel of Experts**, UK Research and Innovation

Current **Elected member, Virology Division Committee**, Microbiology Society

2024 **Chair and organiser, UK ERC Showcase**, Queen's University Belfast
Flagship event with panels featuring Nobel Laureate Prof Emmanuelle Charpentier and Prof Teresa Lambe OBE.

2023 – 2024 **Contributor, US National Academies report "Toward Sequencing and Mapping of RNA Modifications"**, US National Academy of Sciences
Invited ideation panel delegate (2023) and co-author of a commissioned paper (2024).

2021 – 2024 **Grant reviewer**, MSCA (reviewer and rapporteur), US National Science Foundation panel, Swiss National Science Foundation, Polish National Science Centre

Current **Module Coordinator, MSc Research Project**, Wellcome-Wolfson Institute for Experimental Medicine, Queen's

Current **STEM NI Ambassador**, Northern Ireland

2014 – **Fellow of the Winston Churchill Memorial Trust**, Churchill Fellowship, UK

GRANTS AND AWARDS

2021 – 2026 **ERC Starting Grant: PTFLU, post-transcriptional regulation of influenza A virus RNA**, European Research Council · sole PI · grant 948834
€1,568,010 to establish The Courtney Group.

2023 – 2026 **MRC research grant: host ATPases in rhinovirus replication**, UKRI Medical Research Council · co-investigator · MR/X020371/1
£458,574.

2024 – 2025 **Royal Society Partnership Grant**, The Royal Society · with Regent House School
A research project run with secondary school pupils.

2017 – 2020 **Marie Skłodowska-Curie Global Fellowship: IAV-m6A**, European Commission · sole PI · Duke University and Institut Pasteur
€260,000 to study m6A RNA methylation in influenza A virus replication and pathogenesis.

2015 **Postgraduate Student of the Year (John RE Scott Convocation award)**, Ulster University

2014 – 2015 **Churchill Fellowship**, Winston Churchill Memorial Trust
€7,600 for research stays in Italy and Denmark.

PATENT APPLICATIONS

2024 **Novel compounds for the treatment of viral infections**, US patent application 18/550,005

2019 **Compositions and methods for enhanced gene expression and viral replication**, US patent application 16/091,822
Using m6A-associated cellular machinery to boost transgene expression or viral replication.

2019 **Single guide RNA/CRISPR/Cas9 systems, and methods of use thereof**, US patent application 16/326,908
Allele-specific CRISPR/Cas9 systems for treating corneal dystrophies.

SELECTED TALKS AND EVENTS

2026 **Invited speaker: Building ERC-competitive research**, University of Split, Croatia

2024 **Event chair: UK ERC Showcase**, Queen's University Belfast

2023 **Delegate: ideation panel on RNA modification mapping**, US National Academy of Sciences

2020 **Invited speaker: ANRS International Symposium on RNA Modifications**, Institut Cochin, Paris

MENTORING AND LEADERSHIP

- Mentored Dr Hannah Turkington to a competitive three-year BBSRC Postdoctoral Fellowship (2024).
- PhD graduates went straight to postdoctoral posts at the University of Cambridge and the University of Copenhagen.
- MSc graduates have moved into government science roles, including Forensic Science NI.
- EMBO Laboratory Leadership course and UKRI Future Leaders Fellows leadership retreat (2024).

PUBLIC ENGAGEMENT AND MEDIA

- Regular expert voice on BBC television and national radio during the COVID-19 pandemic, explaining viral mutation, immune escape and variant surveillance.
- Long-form articles for The Conversation reaching international readers.
- Works with regional health bodies and schools to promote influenza vaccination awareness.

PROFESSIONAL MEMBERSHIPS

Microbiology Society · American Society for Virology · RNA Society

SELECTED PUBLICATIONS

co-senior author · * co-first author. Full list at davidgcourtney.com/publications and on ORCID.

1. Bonazza S, Turkington HL, Sukumar S, Peate E, Coutts HL, Montgomery JJ, Hawthorn C, Getty EMPE, Touzelet O, Barabas J, Power UF, **Courtney DG** (2026). Myoferlin is a component of late-stage vRNP trafficking vesicles for enveloped RNA viruses. *Nature Communications 17*. Cited 1.
2. Bonazza S, **Courtney DG** (2025). Influenza A virus RNA localisation and the interceding trafficking pathways of the host cell. *PLoS Pathogens 21(4): e1013090*. Cited 10.

3. Bonazza S, Coutts HL, Sukumar S, Turkington HL, **Courtney DG** (2024). Identifying cellular RNA-binding proteins during infection uncovers a role for MKRN2 in influenza mRNA trafficking. *PLoS Pathogens 20(5): e1012231*. Cited 12.
4. Dupont M, Krischuns T, Giai-Gianetto Q, Paisant S, Bonazza S, Brault J, Douché T, Perez-Perri JI, Hentze MW, Cusack S, Matondo M, Isel C, **Courtney DG#**, Naffakh N# (2024). The RBPome of influenza A virus mRNA reveals a role for TDP-43 in viral replication. *Nucleic Acids Research*. Cited 17.
5. Coutts LH, **Courtney DG** (2024). Identifying individual pseudouridine (Ψ) sites across transcripts from HIV-1 infected cells. *HIV Protocols. Methods in Molecular Biology, vol 2807*.
6. **Courtney DG**, Tsai K, Bogerd HP, et al. (2019). Epitranscriptomic regulation of HIV-1 gene expression by m5C. *Cell Host & Microbe 26(2): 217–227*. Cited 227.
7. **Courtney DG**, Chalem A, Bogerd HP, et al. (2019). Extensive epitranscriptomic methylation of A and C residues on murine leukemia virus transcripts enhances viral gene expression. *mBio 10(3): e01209-19*. Cited 77.
8. **Courtney DG**, Kennedy EM, Dumm RE, et al. (2017). Epitranscriptomic enhancement of influenza A virus gene expression and replication. *Cell Host & Microbe 22(3): 377–386*. Cited 228.
9. **Courtney DG**, Moore JE, Atkinson SD, et al. (2015). CRISPR/Cas9 DNA cleavage at SNP-derived PAM enables both in vitro and in vivo KRT12 mutation-specific targeting. *Gene Therapy 23(1): 108–112*. Cited 125.
10. **Courtney DG**, Toftgaard Poulsen E, Kennedy S, et al. (2015). Protein composition of TGFBI-R124C- and TGFBI-R555W-associated aggregates suggests multiple mechanisms leading to lattice and granular corneal dystrophy. *Investigative Ophthalmology & Visual Science 56: 4653–4661*. Cited 38.